

Lecture Notes on

The Search for Extraterrestrial Intelligence

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These notes accompany our second meeting of the year, a guest talk and Q&A with Dr. Seth Shostak, Senior Astronomer at the SETI Institute. SETI stands for the Search for Extraterrestrial Intelligence. This handout is a primer on what that search actually is, how scientists carry it out, and why Dr. Shostak believes we may pick up a signal within our own lifetimes. It is meant to be read before the meeting so the Q&A goes deeper.

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1 What SETI Is, and What It Is Not

SETI is the effort to find evidence of technology built by intelligent life beyond Earth. The key word is *evidence*. SETI does not study UFO sightings or alien visitors. It looks outward, across interstellar distances, for signals or other traces that only a technological civilization could produce.

- **The core assumption.** A civilization with radio or laser technology may either broadcast on purpose or simply leak signals into space, the way our own radar and television have leaked outward for decades.
- **The target.** We are not looking for life in general, such as microbes, but for *intelligent, technological* life. Finding pond scum on Mars would be astrobiology; finding a radio beacon would be SETI.
- **A short history.** Frank Drake ran the first modern search, Project Ozma, in 1960, pointing a radio dish at two nearby stars. The SETI Institute was founded in 1984 and Dr. Shostak has been a leading voice there for decades.

The search rests on a simple bet: in a galaxy of a few hundred billion stars, we are probably not the only ones who learned to build a transmitter.

2 The Drake Equation

To organize the question, Frank Drake wrote down an equation in 1961 that estimates the number of detectable civilizations in our galaxy right now:

$$N = R_{*} \cdot f_p \cdot n_e \cdot f_l \cdot f_i \cdot f_c \cdot L.$$

Each factor narrows the field from all stars down to the ones we could actually hear:

- R_{*} : the rate of star formation in the galaxy, in stars per year.
- f_p : the fraction of those stars that have planets.
- n_e : the average number of potentially habitable planets per such star.
- f_l : the fraction of habitable planets where life actually begins.
- f_i : the fraction of those where life becomes intelligent.
- f_c : the fraction of intelligent species that develop detectable technology.
- L : the average lifetime, in years, that such a civilization keeps broadcasting.

The Drake equation is less a formula for a firm answer than an organized way to list what we do and do not know. The left side of the equation, R_{*} , f_p , and n_e , is now grounded in real data. The right side, especially f_i and L , is still almost pure guesswork, and a single pessimistic factor can drive N down toward one.

A useful takeaway: the term we understand least, L , is also the one we control for ourselves. How long a civilization survives may matter more than how often one appears.

3 How We Actually Search

3.1 Radio SETI and the Narrowband Trick

Most of the search has been done in radio. The reason is a clean physical signature. Natural sources, such as stars and gas clouds, spread their energy across a wide range of frequencies. A deliberate transmitter can instead concentrate power into a very narrow band, close to a single pure tone. Nature does not make pure tones, so a narrowband signal that drifts smoothly with a planet's motion is a strong flag for technology.

3.2 The Water Hole

If you wanted other civilizations to find you, you would broadcast at a frequency everyone can agree on. Radio astronomers point to a quiet band between the neutral hydrogen line at 1.42 GHz and the hydroxyl line at 1.66 GHz. Hydrogen (H) and hydroxyl (OH) together make water, so this band is nicknamed the *water hole*, a natural meeting place for water-based life.

3.3 The Instruments

- **Allen Telescope Array.** A field of 42 radio dishes at Hat Creek, California, built specifically to hunt for signals many hours a day.
- **Optical SETI.** A newer approach that watches for laser pulses lasting only a few billionths of a second, far brighter for that instant than any star.
- **Breakthrough Listen.** A ten-year, one hundred million dollar survey begun in 2015, scanning the one million nearest stars and the hundred nearest galaxies.

3.4 Technosignatures

Beyond deliberate signals, some searches look for *technosignatures*: side effects of a civilization that we could spot at a distance. Examples include waste heat from enormous energy collection, or industrial pollutants in a planet's atmosphere. The signal is not always a message; sometimes it is just the exhaust.

4 The Fermi Paradox

In 1950, over lunch, the physicist Enrico Fermi asked a question that still hangs over the field: *Where is everybody?*

The logic is uncomfortable. The galaxy is roughly 13 billion years old. Even a civilization spreading slowly, star to star, could cross the whole galaxy in a few tens of millions of years, a blink on that timescale. So if intelligent life is common, the galaxy should already be full of it, and we should see the evidence. We do not. That gap between expectation and silence is the Fermi paradox. Common proposed resolutions include:

- **Rare Earth.** Habitable worlds, or life itself, may be far rarer than the raw star count suggests.
- **The Great Filter.** Some hard step, past or future, stops nearly every civilization from spreading.

- **They are quiet.** Civilizations may not broadcast, or may not last long enough to overlap with us.
- **We have barely looked.** The total sky, time, and frequency we have searched is still a tiny fraction of what is possible.

Dr. Shostak often stresses the last point: absence of evidence, after so short a search, is not yet evidence of absence.

5 Why the Numbers Favor a Search

5.1 A Galaxy Full of Worlds

The Milky Way holds a few hundred billion stars. Two decades ago we knew of no planets outside the solar system. Then NASA's Kepler mission changed the picture: planets are the rule, not the exception. A common estimate is that roughly one in five Sun-like stars hosts an Earth-size planet in its habitable zone. That alone implies billions of potentially habitable worlds in our galaxy.

5.2 Moore's Law for the Search

Dr. Shostak's optimism rests less on new physics than on raw computing speed. The rate at which SETI can sift through star systems has roughly doubled every year and a half, tracking the same exponential growth as computer chips. At that pace, the number of systems we can examine over the next two decades is enormous compared with everything searched so far. He has a standing bet, a cup of coffee, that we will detect a signal within about two decades. The claim is not that aliens are certainly there; it is that we are finally searching fast enough to find them if they are.

6 What a Detection Would, and Would Not, Mean

Suppose tomorrow a telescope locks onto a narrowband signal that passes every test. What would we actually have?

- **Proof we are not alone.** The mere existence of the signal, not its content, would be the discovery. It would settle a question humans have asked for millennia.
- **Not a conversation.** The nearest plausible source is many light-years away, so any reply would take years, decades, or longer each way. This is detection, not dialogue.
- **A careful next step.** Scientists follow post-detection protocols: confirm with independent telescopes, rule out human interference, and announce openly rather than secretly.

The value of SETI does not depend on success. Even a long silence teaches us something real about how rare, or how quiet, our neighbors are.

7 Questions to Bring to Dr. Shostak

Come with questions. A few strong ones to start:

- What would convince you, beyond doubt, that a signal was truly extraterrestrial?

- Which Drake equation factor do you think we understand least today?
- Has the explosion of known exoplanets changed how SETI chooses where to point?
- If we found a signal, what is the very first thing the team would do?
- Why radio and lasers? Could an advanced civilization use something we cannot yet detect?

8 Closing

SETI sits at an unusual crossroads of astronomy, biology, engineering, and patience. It asks one of the oldest questions in a form we can finally test with instruments. Whether the answer arrives in Dr. Shostak's two decades or takes far longer, the search itself keeps sharpening the more basic question underneath it: how likely is a universe like ours to produce minds like ours, more than once.